

Test Automation Proposal for

dimagi

Submitted by



Project Overview

Dimagi is seeking to establish a Test Automation Solution for one of its key product lines - CommCare Connect across web and mobile which currently has no automation in place. Primary focus areas are:

- Asses Automation Feasibility
- Design and Implement a robust Test Automation Framework
- CI/CD Integration
- Define an automation roadmap covering functional, regression, and smoke testing
- Enable the internal QA team to maintain and scale automation effectively over time



Objective

The objective of this proposal is to establish a strategic QA partnership between Dimagi and Moolya Software Testing to ensure comprehensive quality ownership and implement a robust test automation solution.

Moolya will leverage its expertise in test automation and quality engineering to enhance product stability, improve testing efficiency, and minimize manual testing efforts. The engagement will focus on:

- Establish a scalable automation framework for both web and mobile platforms.
- Ensure CI/CD readiness through integration with build pipelines.
- Reduce manual effort by automating test flows.
- Deliver measurable outcomes in terms of test coverage, execution efficiency, and release confidence.
- Build automation competency within Dimagi's QA team through knowledge transfer and enablement sessions.

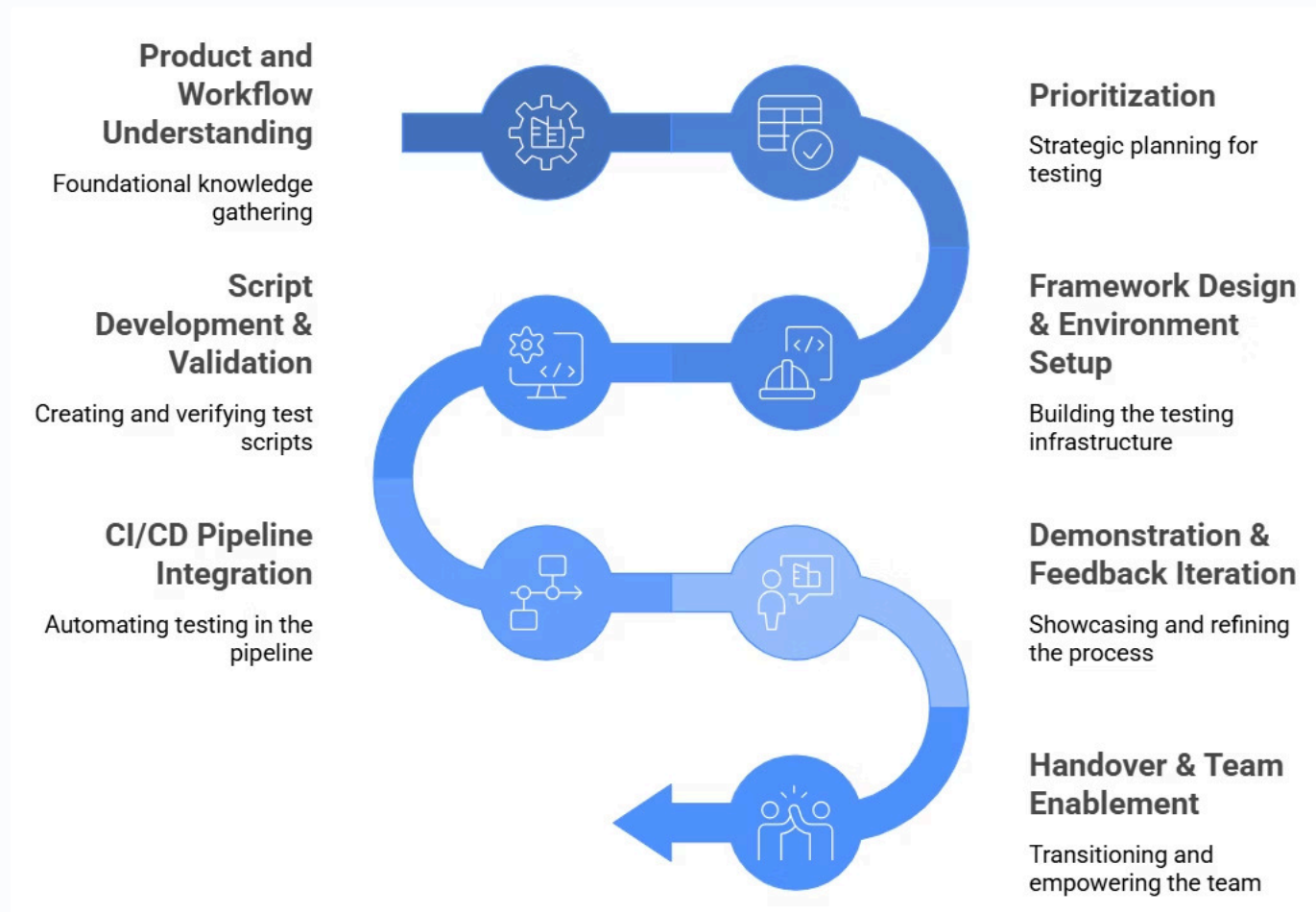


Testing Scope

- Design and implement a scalable automation framework for Dimagi's **CommCare Connect** product covering both **Web** and **Mobile** platforms.
- Collaborate with Dimagi team to prioritize modules and features
- Develop a Test Automation Roadmap defining smoke, feature, and regression test packs, with provisions for future expansion
- Integrate automation with CI/CD pipelines to enable continuous testing
- Implement best practices for framework optimization and long-term sustainability
- Define and track key automation metrics to measure effectiveness
- Upskill the Dimagi QA team to independently maintain, enhance, and scale automation efforts.



Test Automation - Moolya's Approach



Test Execution Plan : 2-month Automation Foundation plan

Month 1 : Discovery, Feasibility & Framework Design

Phase	Activities	Deliverables
Discovery	CommCare Connect Application Understanding	
Assessment & Audit	- Review existing system, process and test environments - Evaluate tools, frameworks, and integration feasibility - Identify automation-ready scenarios for Web and Mobile	Automation Feasibility Report
Test Automation Setup	- Design and implement automation framework - Plan CI/CD integration approach. - Outline roadmap for smoke, regression, and functional automation	- Test Automation Framework - Automation Roadmap
Pilot Automation	- Prioritize key modules - Develop and execute smoke test scripts for key modules - Demonstration of the automated scripts to Dimagi stakeholders	- Test Scripts - Test Execution and Summary Report



Month 2 : Automation Framework Optimization, Best Practices and Knowledge Transfer

Phase	Activities	Deliverables
Continued Automation Execution	• Develop and execute next set of smoke test scripts • Integration with CI/CD pipeline • Refine the roadmap and strategy for smoke, regression, and functional automation	• Automated Smoke Test Scripts (Web and Mobile) • CI/CD-enabled test execution setup • Test Execution and Summary Report
Optimization and Best Practices	• Optimization of the test automation framework • Refine scripts for maintainability and modularity • Finalize metrics to track automation efficiency • Documentation of automation process and maintenance guide • Demonstration of the automated scripts to stakeholders	• Optimized Automation Framework • Test Scripts • Automation Metrics Defined
Knowledge Transfer and Handover	• Conduct QA enablement and training sessions to Dimagi team • Share automation process and maintenance document • Handover framework ownership	• Knowledge transfer completion • Automation documentation and maintenance guide • QA team ready for independent ownership



Requirements

Manage every requirement, map to test cases, monitor coverage, and keep traceability through every sprint.

Sort

Folders

All

One Way Sync (64)

Monitor Branch Changes

Monitor Pull Request (3)

Monitor Push Request

Mapping Bugasura Issues

Status Changes

Integration Workflow

Sprints v2.1

+ Add

#REQ01

A valid credit card is available and navigation to the payment page is successful.

5

#REQ02

Resend OTP Functionality

6

#REQ05

OTP autofill on receipt

2

#REQ06

Fallback mechanism activation when all gateways are done

0

Test Repo

Create, group, and manage test cases the way your team works – not the other way around.

Sort

Folders

All

One Way Sync (64)

Monitor Branch Changes

Monitor Pull Request (3)

Monitor Push Request

Mapping Bugasura Issues

Status Changes

Integration Workflow

Debug Mode

Get Repositories

Installation Callback

Monitor Pull Request

+ Add Test Case

#TS01

A valid credit card is available and navigation to the payment page is successful.

LOW

#TS02

Resend OTP Functionality

HIGH

#TS05

OTP autofill on receipt

CRITICAL

#TS06

Fallback mechanism activation when all gateways are done

LOW

#TS07

Integrating with external tools

HIGH

#TS08

Validate the error message when user enter invalid phone number and remaining fields are valid

CRITICAL

Traceability Matrix

RE001

User Authentication & Access Control

RE002

Subscription & Billing Management

RE003

API & Integrations

RE004

Multi-Tenancy Support

	Syed	Madhan	Anbu	Kunwar Gyaneshwar	Arjun	Soura
Test Case	Windows 11	Windows 12	iOS	iPhone 18pro	iPhone 16	Andro
TC001 Verify that users can register and log in using valid credentials.	12	2	2	3	3	
TC002 Ensure multi-factor authentication (MFA) is triggered when enabled.		2		3		
TC003 Test login attempts with incorrect passwords to enforce account logout.	4		2	3	1	
TC004 Validate that users with different roles have appropriate access permissions.		4	2	3	2	
TC005 Test auto-renewal functionality for recurring subscriptions.	2		3	3	2	

RE005

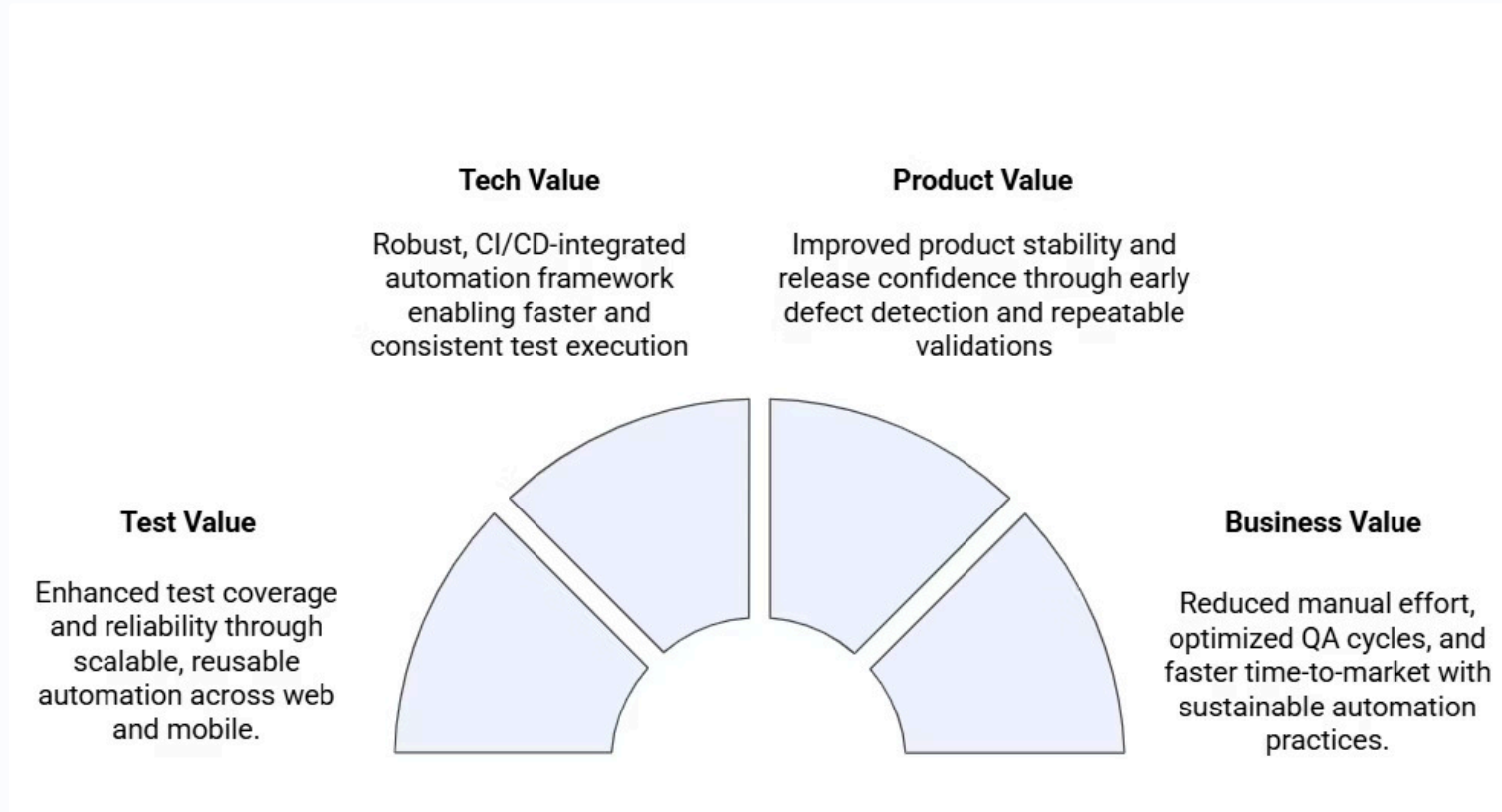
Performance & Scalability

Deliverables

- Test Plan and Strategy Document with Automation Roadmap
- Test Automation Framework
- Automated Test Scripts
- Test Summary Report with execution status and insights
- Script Maintenance Document
- Knowledge Transfer and Training Document



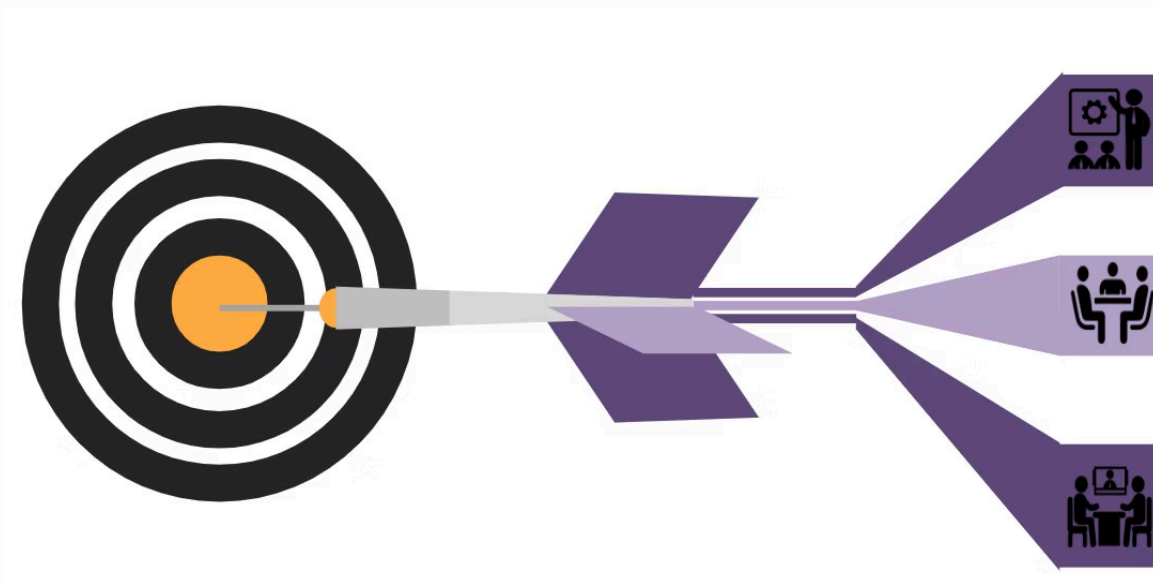
Value Delivered



Our Commitment

We will setup a dedicated testing team supported by Domain expert and UX experts, to conduct comprehensive end-to-end testing of Instacoins' web application across both Flights and Stays modules.

Unified Expertise with Moolya's 3-in-a-Box Model



Combination of Expertise: Integrated Under One Roof

By combining Delivery Engineering, Testing Experts, and Domain SMEs into one cohesive unit, Moolya offers a comprehensive, cost-efficient solution that drives faster delivery, higher quality, and scalable results

Delivery Engineering Team

Bringing Technical Excellence through Delivery Engineering

- Collaborative Development
- Custom Integrations
- Continuous Delivery Focus

Testing Experts

Expert QA Testing to Ensure Compliance and Quality

- Context Driven Testing
- Automated Testing Frameworks
- End-to-End Testing

Domain Subject Matter Expert (SME) Team

Domain Experts Ensuring Business and Project Alignment

- Deep Domain Knowledge
- Technical Guidance & Expertise
- Collaboration

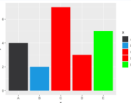


Assumptions

- **Product Overview:** The Dimagi team will provide Moolya with a comprehensive walkthrough of the framework.
- **Point of Contact:** A designated point of contact will be available for the Moolya team to reach out to for assistance.
- **Requirement Stability :** All functionalities will be clearly defined, and remain stable throughout the project timeline.
- **Environment Availability :** A dedicated and stable testing environment will be available for the Moolya team.
- **Access to Test Data :** Necessary test data will be provided by the Dimagi team.
- **Documents :** Dimagi will provide the necessary documents required for the engagement.



Quality Metrics

	Metric	Impact
	Test Coverage (Percentage of tests covered by automation)	- Ensures better regression testing and reduces production defects.
	Defect Detection Rate (Rate at which automated tests identify defects)	- High rate ensures early defect identification, improving software quality. - Reduces debugging efforts in later stages of development.
	Test Execution Time (Average time taken for automated test suites to run)	- Faster execution speeds up feedback loops, reducing development cycle time. - Long execution times slow down releases and CI/CD processes.
	Automation Pass Percentage (Percentage of tests passing)	- High pass percentage indicates system stability. - Low percentage may signal frequent failures, requiring deeper analysis.
	Flaky Tests (Percentage of unstable or inconsistent tests)	- High flakiness reduces trust in automation. - Leads to manual verification efforts and increased debugging time. - Stable tests improve efficiency and confidence in results.
	Stability Index Percentage (Percentage of builds passing)	- High stability index indicates a reliable and stable system. - Frequent build failures disrupt development and delay releases.

Pricing

Engagement Plan	Duration	Monthly Cost (INR)
Automation Foundation Plan	2 Months	₹4,00,000 / month

Payment Terms

- Payment to be done within 30 days of invoice generation
- Any applicable taxes will be charged extra on the pricing mentioned
- Payment Milestone 1: 50% of the total project cost shall be payable upon signing the Statement of Work (SoW), as confirmation of project initiation.
- Payment Milestone 2: Remaining 50% of the invoice amount on completion of one month from the date of submission of the initial test report or immediately upon completion of the retest, whichever is earlier.
- As part of our standard operating procedures, Moolya ensures comprehensive insurance coverage including liability, security, cybersecurity, and employee welfare. Should there arise a necessity for project-specific insurance, such coverage will be procured and billed accordingly as a pass-through expense

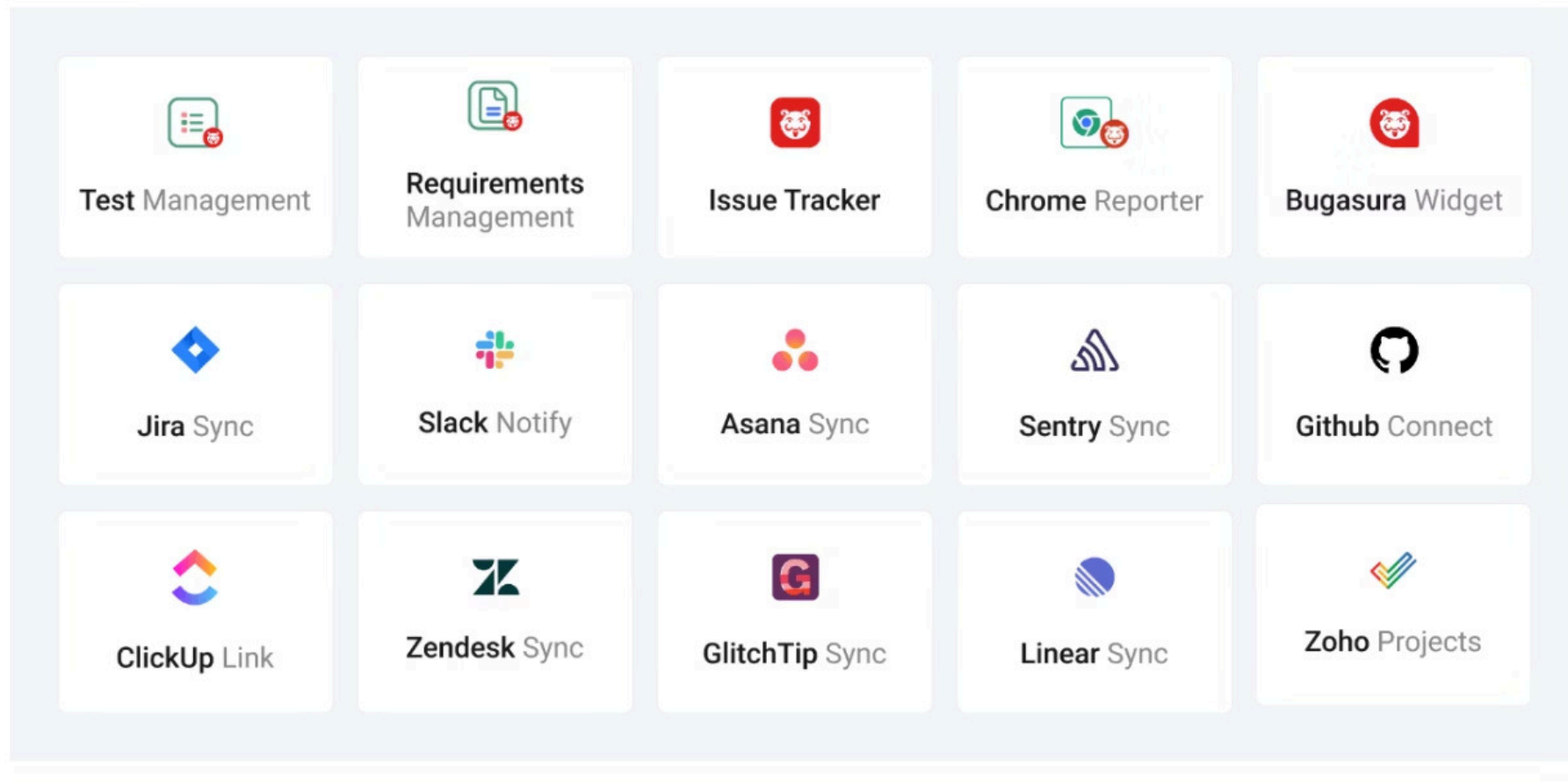
Important Note: Preliminary Estimate and Scope Adaptability

This proposal has been prepared based on our current, high-level understanding of Dimagi's outlined requirements. It serves as an initial estimate, providing a foundational overview of the proposed testing strategy, phases, and deliverables.

We acknowledge that the complexities of the application, along with specific testing needs, may evolve as we progress further into the project. Therefore, we anticipate refining this proposal to align more closely with the project's specific requirements and objectives, guided by our collaborative engagement with the Dimagi team.



Bugasura Capabilities



Thank You!

